EXPERIMENT 2

Logical, Bit-Manipulation and Shift Operations Using 8-bit and 16-bit Data Manipulation

a) Verification of Bit Masking Operations Using 8-Bit Processor Assembly Language Programming

b) Testing of Bit Setting and Clearing Operations Using 8-Bit Assembly Language Programming

Objective:

To develop and verify **logical, bit-manipulation, bit setting, and bit clearing operations** on 8-bit data using **8085 microprocessor assembly language programming**.

Algorithm:

- Initialize the required memory locations and registers.
- Load the 8-bit input data into the appropriate register.
- Perform **bit masking** using logical operations such as ANA to mask the required bits.
- Perform **bit setting** by using the appropriate logical operation to set the selected bit(s) to 1.
- Perform **bit clearing** by using the appropriate logical operation to clear the selected bit(s) to 0.
- Store the obtained results in the specified memory locations.
- Verify the results with the expected bit-manipulation values.

a) Bit Masking Operations:

Code:

```
LDA 4500h
ANI 0Fh
STA 4502h
HLT
```

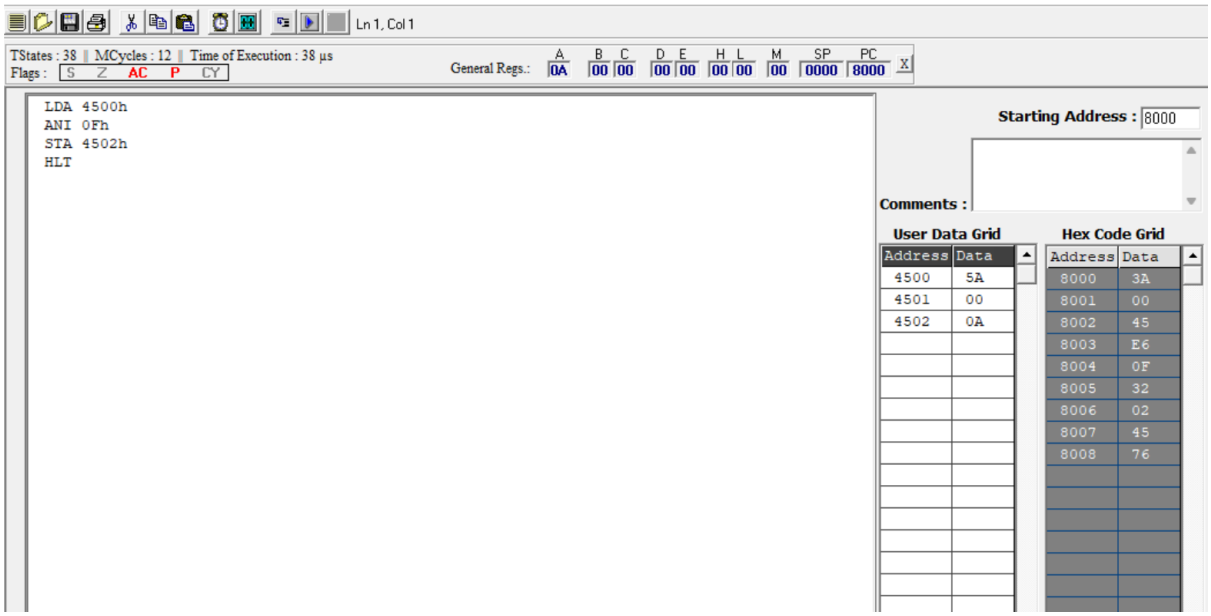
Input:

Input Address	Data
4500h	5Ah

Output:

Output Address	Data
4501h	00h
4502h	0Ah

Output Screenshort:



(ii) Bit Setting and Clearing Operations:

Code:

```
LDA 4500h
ORI 08h
STA 4501h
LDA 4501h
ANI F7h
STA 4502h
HLT
```

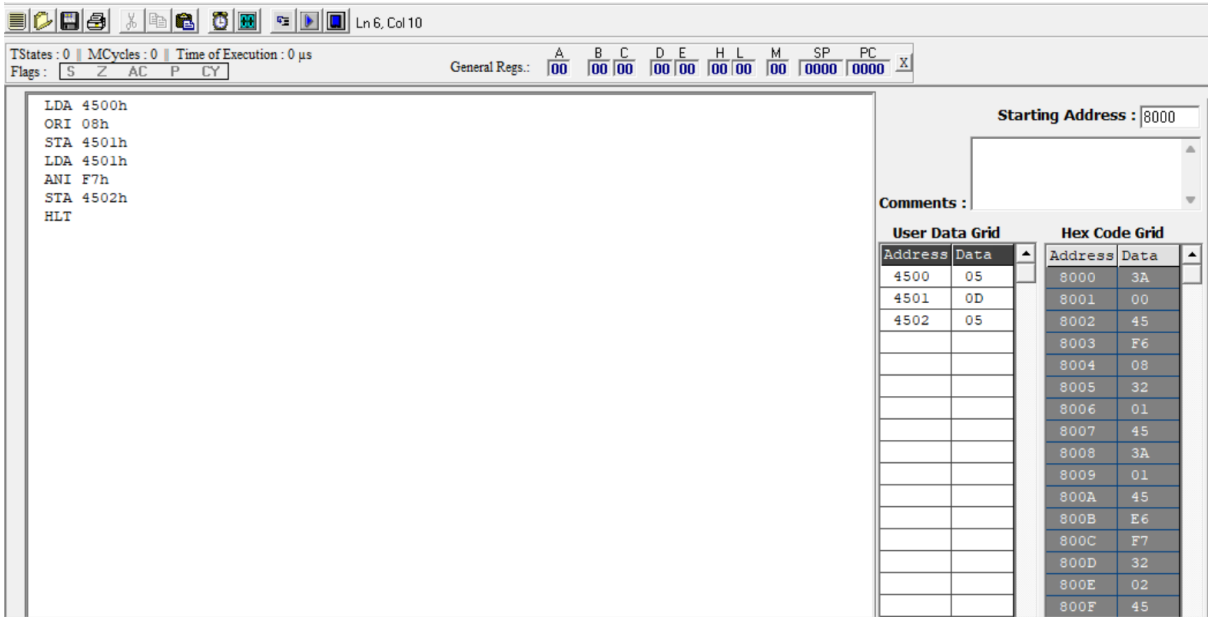
Input:

Input Address	Data
4500h	05h

Output:

Output Address	Data
4501h	0Dh
4502h	05h

Output Screenshort:



Result:

Thus, the **bit masking, bit setting, and bit clearing operations** were successfully developed and verified using **8085 8-bit microprocessor assembly language programming**. The expected results were obtained correctly.

Student Reflection:

I learned how to perform **bit-level manipulation of 8-bit data** using 8085 assembly language. I understood how logical operations can be used for **masking, setting, and clearing individual bits**. This experiment improved my understanding of binary data, logical instructions, registers, and practical 8085 programming.